

Spencer Wallace

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EDUCATION

University of Washington

PhD Astronomy

Seattle, Washington

Oct 2015 – August 2023

University of Arizona

BS Computer Science, Astronomy and Physics

Tucson, Arizona

Aug 2009 – May 2014

WORK EXPERIENCE

Postdoctoral Researcher

University of Washington, Astronomy Department

September 2023 – Present

- Applied C++/CUDA to accelerate large-scale astrophysics simulations on NVIDIA Grace Hopper hardware, analyzing performance data to guide optimization, improving runtime scalability of our scientific HPC code on up to 500+ GPUs and 30,000+ cores
- Ran and analyzed large-scale simulations to verify the correctness and performance of the code, ensuring robust and reproducible results for users of the code base
- Collaborated with HPC specialists across physics, CS, and biology to adapt algorithms adapt algorithms for diverse data-heavy research problems

Research Assistant

University of Washington, Astronomy Department

October 2019 – August 2023

- Built a PyTorch-based generative ML pipeline to synthesize initial conditions for astrophysical simulations, reducing compute time requirements by 900k+ CPU hours
- Developed scalable algorithms for collision detection and growth of solid bodies, enabling the first-ever simulations of planet formation with realistic-sized objects
- Analyzed large simulation datasets using pandas/NumPy, producing insights published in 3 first-author papers
- Built a time-series analysis pipeline using a gaussian process model to find over 7,000 new flare stars in NASA space telescope data
- Mentored undergraduates on Python-based data pipelines and visualization

Teaching Assistant

University of Washington, Astronomy Department

October 2015 – September 2019

- Led weekly discussion sections, graded exams, designed homework assignments and performed guest lectures for undergraduate astronomy classes, some of which involved teaching basic data analysis and numerical modeling skills using Python and Excel
- Collaborated with a team of other teaching assistants to ensure exams and assignments were graded consistently and fairly

Software Consultant

LSST Corporation

June 2011 – August 2014

- Designed interactive visualization widgets (JavaScript/Node.js) for [ASCOT](#) the astronomer's collaborative toolkit
- Built a fully customizable, persistent dashboard interface, along with documentation to facilitate large survey data exploration

SKILLS

Programming Languages: C++, Python, Bash

Tools and Environments: Git, NumPy, pandas, PyTorch, SLURM, matplotlib, seaborn

Communication: 3 first-authored publications, 2 co-authored publications, 8 conference talks, 3 conference posters, 20 pop-sci articles published through [astrobites](#)

PUBLICATIONS

Wallace, S., Quinn, T., "Planetesimal Accretion at Short Orbital Periods", [ApJ](#), 954:61, (2023).

Wallace, S., Quinn, T., Boley, A., "Collision rates of planetesimals near mean-motion resonances", [MNRAS](#), 503(4):5409–5424, (2021).

Wallace, S. and Quinn, T., "N-body simulations of terrestrial planet growth with resonant dynamical friction", [MNRAS](#), 489(2):2159–2176, (2019).

Kim, J., Agertz, O., Teyssier, R., Butler, M., Ceverino, D., Choi, J., Feldmann, R., Keller, B., Lupi, A., Quinn, T., Revaz, Y., Wallace, S (and 31 more), "The AGORA High-resolution Galaxy Simulations Comparison Project. II. Isolated Disk Test", [ApJ](#), 833(2):202, (2016).

CONFERENCE PRESENTATIONS

2nd N-body Shop Collaboration Meeting, **Contributed Talk**, August 2025

24th Charm++ Workshop, **Contributed Talk**, April 2024

241st AAS Meeting, **Dissertation Talk**, January 2023

Exoplanets in Our Backyard II, **Poster**, November 2022

2nd N-body Shop Collaboration Meeting, **Contributed Talk**, June 2022

53rd AAS Division for Dynamical Astronomy Meeting, **Contributed Talk**, May 2022

TESS Science Conference II, **Poster**, August 2021

Sagan Exoplanet Summer Workshop, **Poster**, July 2021

52nd AAS Division for Dynamical Astronomy Meeting, **Contributed Talk**, May 2021

1st N-body Shop Collaboration Meeting, **Contributed Talk**, January 2021

50th AAS Division for Dynamical Astronomy Meeting, **Contributed Talk**, June 2019

49th AAS Division for Dynamical Astronomy Meeting, **Contributed Talk**, April 2018